

then runs on a router that announces its connected IP networks to the other ZeroTier routers. The outcome resembles a multipoint VPN that connects all locations and is managed from a central location. The advantage of this solution is that it eliminates the manual configuration that is typical for OpenVPN and IPsec.

The best operating system for the new ZeroTier router is a Linux/UNIX distribution that focuses on routing and firewalls, such as OpenWrt, VyOS, and OPNsense. The ZeroTier software is installed via a package manager and requires the well-known network ID from the previous sections. After authorisation in ZeroTier Central, the new routers are regular members of the ZeroTier group.

The routers now recognise each other, but they do not yet know all the attached local networks. There are several possibilities for exchanging routing information:

- The routers learn the path information to the other locations from the ZeroTier server. For this to work, enter the IP network in the field 'Managed Routes' of ZeroTier's Web interface (Figure 6). The ZeroTier clients will learn from this central routing table upon connection. This method is suitable for all environments as long as all routers have a connection to ZeroTier. In the free version of ZeroTier, the maximum number of routes is 32.
- Alternatively, the routers use a dynamic routing protocol, such as OSPF. The routers establish neighbourships through the ZeroTier network and use them to exchange IP information. This method is suitable for large environments or if some routers do not have a connection to ZeroTier.

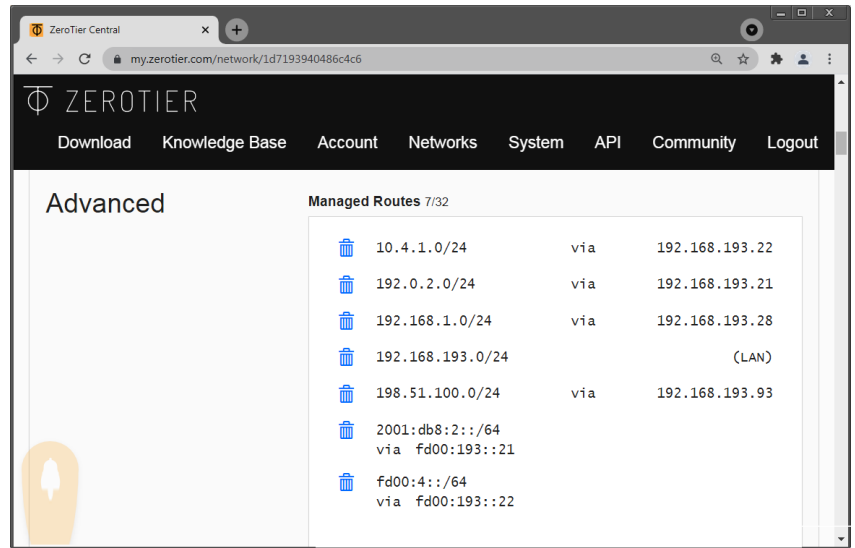


Figure 6: ZeroTier manages and distributes IP routes to its clients

Outlook

The design of ZeroTier's VPN is similar to a software-defined wide-area network (SD-WAN). In fact, ZeroTier with its central routing table, flow rules, and multipath has many similarities to SD-WAN. However, ZeroTier's VPN is missing a zero-touch deployment of routers and granular control over which traffic streams should flow over which VPN connection.

Name resolution of ZeroTier devices was a weak point for a long time, so users mostly had to work with static IP addresses. As a workaround, there are community projects, like ZeroDNS, and finally, ZeroTier provided the official DNS service 'ZeroNSD' — but only in the beta stage.

Those who find the 'Key Networks' Web interface lacking functionality, can switch to the full-blown software ZeroUI. The software has a large number of dependencies and only runs in a Docker container. After a rather

complex installation, a full-fledged Web UI is available that implements virtually all of ZeroTier's functions.

Summary

ZeroTier is a VPN solution that is easy to configure and is usable on all major operating systems and smartphones. The 'ZeroTier One' client automatically negotiates the tunnels, algorithms, and keys. To access the global network, the client only needs a network ID, which the administrator generates after registering in ZeroTier's Web portal. As soon as all the clients are registered, they can reach each other and find the shortest paths between them.

The initial contact of the ZeroTier client is always against the provider's root servers. An Internet connection is mandatory, even if the devices are located in the same network segment. This dependency is compensated by simple configuration, great flexibility, and high security. **END** 🐧

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